Rong Ni

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Education

University of Chicago Sep. 2025 – Present
Master of Science in Computer Science

Xiamen University Sep. 2021 – Aug. 2025
Bachelor of Engineering in Digital Media Technology; GPA: 3.53/4.00

Technical Skills

Languages: Python, C#, C++, JavaScript, SQL
Frameworks / Tools: PyTorch, React, Electron, Node.js, Unity, SQLite, Git, Maya
AI / ML: LLM application development, prompt engineering, structured output validation, GANs

Experience

Human-Computer Integration Lab, University of Chicago Jan. 2026 – Mar. 2026
Research Practicum

- Developed VR demoscene environments for user studies for *Electrotactile Improves Thermal Referral*, a manuscript submitted to ACM UIST 2026 and currently under review; second author.
- Contributed experimental photography used in the manuscript.

Mocap Steganography with GANs Sep. 2023 – May 2024
Research Assistant, Xiamen University

- Adapted SteganoGAN from image steganography to BVH motion-capture data by building a preprocessing pipeline that converted BVH sequences into 3-channel tensors for convolutional training.
- Trained and evaluated Basic, Dense, and Residual encoder-decoder-critic variants in PyTorch on the Bandai-Namco MotionDataset; achieved 99.3% character accuracy when recovering about 7,000-bit payloads while preserving visually indistinguishable motion.
- Analyzed fidelity-recovery trade-offs and robustness under Gaussian noise to compare model behavior across payload accuracy and motion distortion.

Projects

Paper Reader | *Electron, React, Node.js, SQLite, LLM APIs*

- Built a local-first desktop research-paper reader with PDF import, annotations, structured summaries, paper chat, term lookup, contextual translation, and editable Markdown note generation.
- Engineered multi-provider LLM execution across 8 configurable providers with timeout, retry, backoff, and fallback handling for transient failures.
- Implemented prompt contracts, structured-output validation and repair, source-grounded QA, long-text chunking, and response caching for reliable research-paper workflows.

Unity Development Projects | *Unity, C#, Emerald AI, AR Foundation*

- **Fantastic Worriers:** Built combat systems with C#, Emerald AI, Animator state transitions, collision-based damage, a two-phase boss fight, and Animation Event-driven combat audio.
- **Guardians of Chronos:** Built a sci-fi combat level with scene layout, trigger-based progression, spawn logic, UI, win/loss conditions, and NavMeshAgent enemy patrol navigation.
- **Little Prince:** Created AR narrative interactions with AR Foundation image tracking, ARTrackedImageManager, and event-driven dialogue UI for the Drunkard and Lamplighter scenes.

Awards

University Academic Distinction Award | *Top 10%* 2022, 2023, 2024